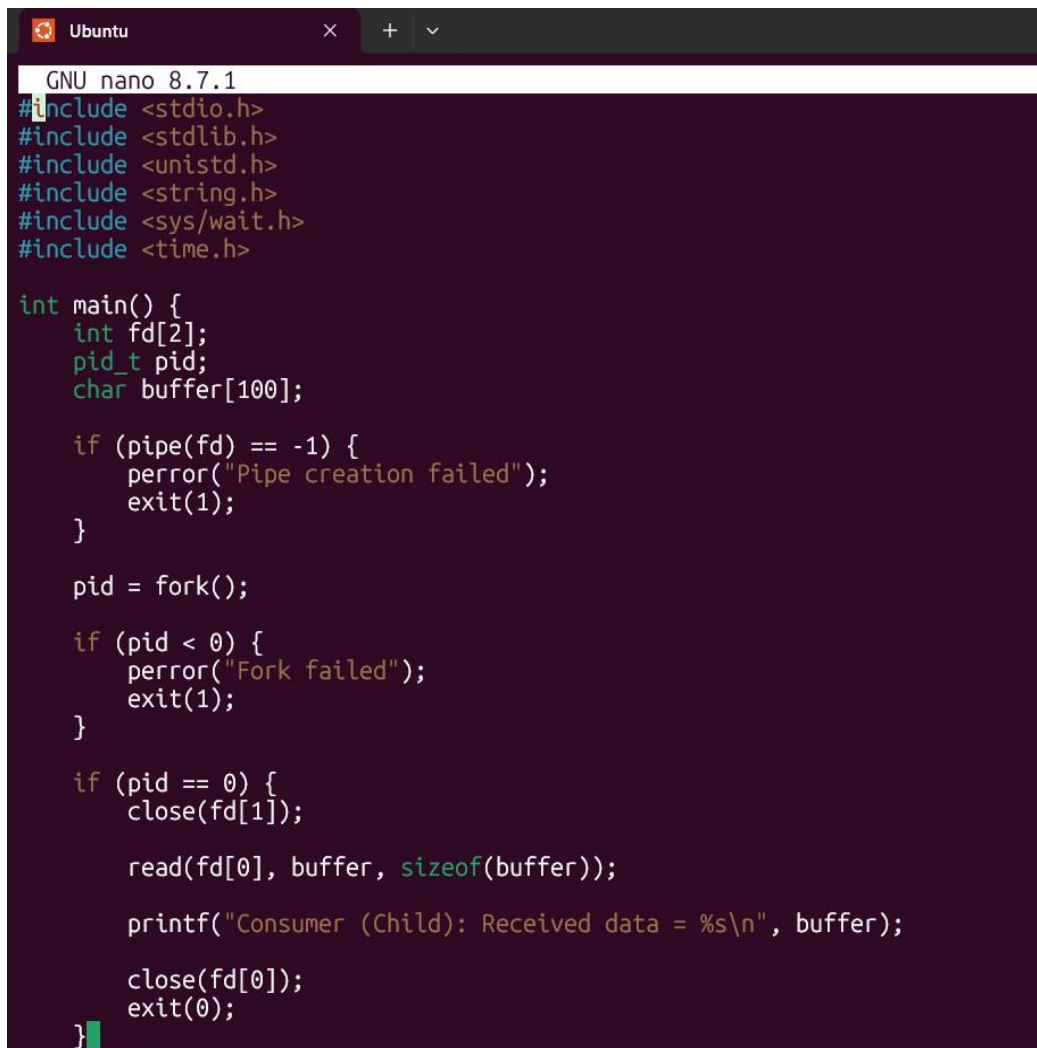


Practical Week 5

Task 1:

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.



```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>
#include <time.h>

int main() {
    int fd[2];
    pid_t pid;
    char buffer[100];

    if (pipe(fd) == -1) {
        perror("Pipe creation failed");
        exit(1);
    }

    pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        exit(1);
    }

    if (pid == 0) {
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Consumer (Child): Received data = %s\n", buffer);

        close(fd[0]);
        exit(0);
    }
```

```

    else {
        close(fd[0]);

        strcpy(buffer, "Hello from Producer!");
        printf("Producer (Parent): Sending data = %s\n", buffer);

        int data_size = strlen(buffer) + 1;

        clock_t start = clock();

        write(fd[1], buffer, data_size);

        clock_t end = clock();

        double communication_time =
            (double)(end - start) / CLOCKS_PER_SEC;

        printf("Data size: %d bytes\n", data_size);
        printf("Communication time: %f seconds\n",
            communication_time);

        if (communication_time > 0) {
            double throughput =
                data_size / communication_time;

            printf("Communication throughput: %f bytes/second\n",
                throughput);
        }

        close(fd[1]);

        wait(NULL);

        printf("Producer: Consumer process completed.\n");
    }
}

```

Output:

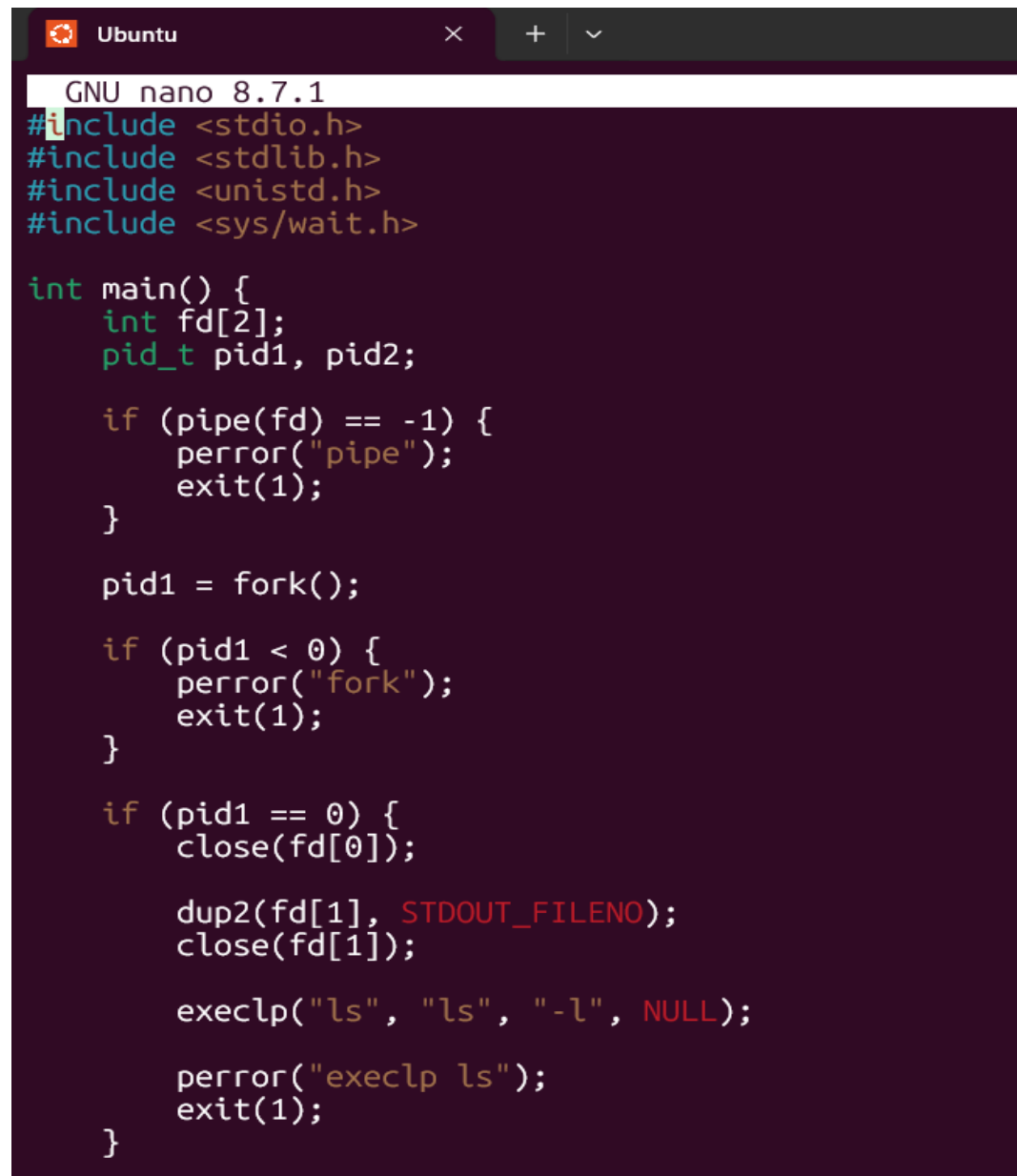
```

nocturnal@nocturnal:~$ nano pipe.c
nocturnal@nocturnal:~$ gcc pipe.c -o pipe
nocturnal@nocturnal:~$ ./pipe
Producer (Parent): Sending data = Hello from Producer!
Data size: 21 bytes
Communication time: 0.000003 seconds
Communication throughput: 7000000.000000 bytes/second
Consumer (Child): Received data = Hello from Producer!
Producer: Consumer process completed.
nocturnal@nocturnal:~$

```

Task-2

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

A screenshot of a terminal window with a dark purple background. The window title bar shows the Ubuntu logo and the text 'Ubuntu'. Below the title bar, the text 'GNU nano 8.7.1' is displayed. The main area of the window contains C code written in a monospaced font with syntax highlighting. The code implements a program that creates a pipe, forks a child process, and executes 'ls -l' in the child, filtering for files ending in '.c' using 'grep'.

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pid_t pid1, pid2;

    if (pipe(fd) == -1) {
        perror("pipe");
        exit(1);
    }

    pid1 = fork();

    if (pid1 < 0) {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0) {
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp ls");
        exit(1);
    }
}
```

```

pid2 = fork();

if (pid2 < 0) {
    perror("fork");
    exit(1);
}

if (pid2 == 0) {
    close(fd[1]);

    dup2(fd[0], STDIN_FILENO);
    close(fd[0]);

    execlp("grep", "grep", ".c", NULL);

    perror("execlp grep");
    exit(1);
}

close(fd[0]);
close(fd[1]);

waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);

return 0;
}

```

Output:

```

nocturnal@nocturnal:~$ nano task2.c
nocturnal@nocturnal:~$ gcc task2.c -o task2
nocturnal@nocturnal:~$ ./task2
drwxrwxrwx 6 nocturnal nocturnal 4096 Aug 14 14:35 OSSP_2520090085
-rwxr-xr-x 1 nocturnal nocturnal 16464 Sep 12 03:44 pipe
-rw-r--r-- 1 nocturnal nocturnal 1457 Sep 12 03:44 pipe.c
-rwxr-xr-x 1 nocturnal nocturnal 16288 Sep 12 03:21 task1
-rw-r--r-- 1 nocturnal nocturnal 903 Sep 12 03:21 task1.c
-rwxr-xr-x 1 nocturnal nocturnal 16312 Sep 12 03:24 task1_waitpid
-rw-r--r-- 1 nocturnal nocturnal 903 Sep 12 03:24 task1_waitpid.c
-rwxr-xr-x 1 nocturnal nocturnal 16296 Sep 12 03:46 task2
-rw-r--r-- 1 nocturnal nocturnal 892 Sep 12 03:45 task2.c
-rwxr-xr-x 1 nocturnal nocturnal 16208 Sep 12 03:36 zombie
-rw-r--r-- 1 nocturnal nocturnal 684 Sep 12 03:35 zombie.c
nocturnal@nocturnal:~$

```